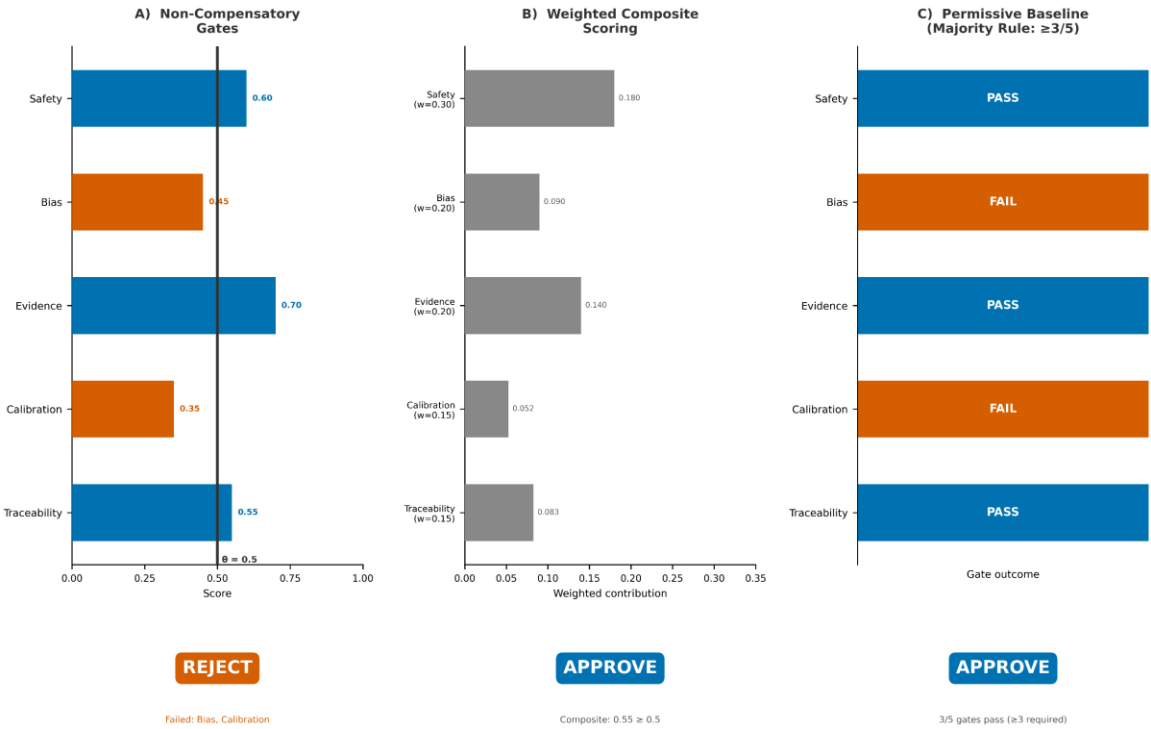


Same evidence profile → three different deployment outcomes

Hypothetical profile: Safety=0.60, Bias=0.45, Evidence=0.70, Calibration=0.35, Traceability=0.55



Supplementary Materials

Companion to Paper 2 — JMIR Medical Informatics

Supplementary Appendix A: Reproducible Evidence Enrichment and Governance Pipeline

This pipeline is described in full in Supplementary Appendix A of Paper 1. A summary is provided here for context.

The evidence base was generated through a four-stage computational pipeline with cryptographic integrity verification: (1) multi-source evidence retrieval for 39 cases; (2) confidence-weighted triangulation yielding 76 features across 37 cases; (3) canonicalisation into a versioned dataset (v1.0.0); and (4) repository integration (v3.1.0-eee) with manifest-based verification (106 files, SHA-256, 30/30 validation checks). The pipeline is deterministic: re-execution from identical source data produces identical outputs.

Supplementary Appendix A (continued): Enriched Triangulation Evidence

This pipeline is described in full in Supplementary Appendix A of Paper 1. A summary is provided here for context.

A subsequent evidence enrichment and triangulation pipeline processed 39 cases through multi-source evidence retrieval, yielding 37 fully triangulated cases and 2 blocked cases (insufficient admissible evidence). Across the 37 processed cases, triangulated confidence scores ranged from 0.73 to 0.93, with 21 cases achieving STRONG triangulation integrity and 16 classified as FRAGILE. Evidence grades were distributed as follows: 26 IDEAL, 5 ACCEPTABLE, 6 MINIMUM, and 2 INSUFFICIENT. The two blocked cases (H34: Acclarent TruDi Navigation System; H35: Sonio Detect) were retained as monitoring-only entries with no feature values assigned, consistent with the framework's conservative treatment of insufficient evidence. All enrichment artefacts are archived in the Ethical Alpha Audit Core Scientific Record (Zenodo DOI: to be provided upon publication) with SHA-256 hash-chain verification.

Supplementary Appendix B: PhysioNet Empirical Heterogeneity Test

The simulation's central testable prediction — that governance evidence profiles in practice exhibit heterogeneous patterns where AI systems score differently across governance domains — was tested using governance vectors derived from two sepsis prediction models evaluated on PhysioNet 2019 clinical data (5,000 patients, 188,453 patient-hours). Both models exhibited heterogeneous evidence profiles (feature spread > 0.30; GradientBoosting spread = 0.474, RandomForest spread = 0.468). The GradientBoosting model scored strongly on bias (0.912 normalised) but weakly on safety (0.438), while the RandomForest model showed the reverse pattern on calibration (0.886) versus safety (0.418). This cross-domain variation — where strengths in one governance dimension coexist with weaknesses in another — matches the heterogeneous evidence model under which the simulation demonstrated the strongest gate safety advantage over composite scoring. Ranking stability analysis confirmed that model governance rankings were stable under 90% of scoring weight perturbations ($\pm 20\%$), with rank changes occurring only when bias or calibration weights were substantially altered. These empirical findings are consistent with, though not a formal validation of, the simulation's prediction that conjunctive gate logic provides structural protection specifically because real governance evidence is heterogeneous.

Supplementary Appendix C: Extended Scope Conditions

This analysis is subject to several limitations. Retrospective evaluation relies on documented failure cases and may not capture unknown failure modes. Simulation-based findings depend on modelled

assumptions. Operational feasibility estimates are not derived from prospective deployment. Accordingly, findings should be interpreted within these constraints.

This study contributes to a coordinated set of analyses examining a non-compensatory evaluation approach across complementary domains, including formal logic, simulation behaviour, parameter specification, retrospective validation, and deployment trade-offs.

Estimated execution times (45–90 minutes for primary audit; 10–25 minutes for secondary validation) are consistent with existing governance activities such as multidisciplinary review processes. These estimates are derived from structured task decomposition and should be interpreted as indicative.

In retrospective evaluation, a subset of documented failure cases satisfied conventional performance-based criteria despite exhibiting clinically material failure modes. Under the evaluated non-compensatory approach, these cases did not meet minimum domain-specific requirements, indicating potential limitations of compensatory evaluation frameworks.

All analyses were conducted using deterministic computational procedures with version-controlled inputs and outputs. Where applicable, artefacts were linked via cryptographic hashing to enable independent verification of result provenance. This design allows verification that outputs correspond to specific, fixed inputs without post hoc modification.

Under uniform or random failure distributions, non-compensatory gating may not outperform weighted models, highlighting defined boundary conditions for applicability.

Simulation-informed estimates indicate that Primary Safety Thresholds constitute a minority of total thresholds (<10%), with most decisions operating under a reduced three-layer documentation requirement. Estimated completion times range from 45–90 minutes (Primary) and 10–25 minutes (Secondary).

Supplementary Appendix D: NHS Governance Artefact Mapping

Mapping simulation parameters to NHS governance artefacts

To anchor the simulation parameters in recognisable institutional contexts, the five governance domains correspond illustratively to the following NHS governance artefacts: Gate 1 (Safety) maps to the NICE Evidence Standards Framework Tier 3b evidence requirements and MHRA clinical safety case documentation; Gate 2 (Equity) maps to the NHS AI and Digital Regulations Service (AIDRS) bias-assessment requirements and the Equality Impact Assessment mandated under the Equality Act 2010; Gate 3 (Documentation) maps to the DCB0129 clinical safety case report, model card requirements, and DTAC transparency criterion; Gate 4 (Accountability) maps to the NHS Clinical AI Oversight Committee (CAIOC) terms of reference, Caldicott Guardian function, and Senior Information Risk Owner (SIRO) designation; and Gate 5 (Monitoring) maps to the post-market surveillance obligations under MHRA guidance and the NHS Federated Data Platform monitoring capabilities. This mapping is illustrative rather than definitive; it indicates that the simulation's abstract parameters have concrete institutional referents in NHS governance practice without implying that these specific artefacts are the only valid operationalisations.

Primary Evidence Model: Heterogeneous Profiles

The primary model specifies that unsafe tools exhibit heterogeneous evidence profiles: scoring strongly in visible governance domains (vendor-reported safety metrics, documentation) and poorly in latent domains (equity subgroup analysis, monitoring infrastructure). This is the condition under which composite scoring's compensation mechanism is theoretically exploitable, and therefore the condition under which a test of gate logic is most theoretically informative. This simulation evaluates gate performance under a specific, theory-motivated evidence heterogeneity pattern; it does not claim universality across all possible governance evidence distributions.

Portfolio Size Justification

These parameters are explicitly modelled rather than empirically derived from an observed institutional dataset; the sensitivity analyses below test robustness to their specific values. A portfolio of 1,000 simulated AI tools was selected for three reasons. First, statistical power: at the expected unsafe base rate (~20% of tools being genuinely unsafe), 1,000 tools provide sufficient precision to detect unsafe deployment rates at the 0.5–1.0% level with bootstrap 95% confidence interval half-widths of approximately ± 0.5 percentage points. Second, stability: preliminary runs at $N=100$, $N=500$, and $N=1,000$ confirmed that point estimates stabilised by $N=500$ and confidence intervals narrowed further at $N=1,000$ without qualitative change in findings. Third, plausibility: while no single NHS trust manages 1,000 AI tools, the figure approximates a system-level portfolio for a national health system or regional consortium, which is the governance scale at which aggregate safety properties become most policy-relevant [3].

Supplementary Appendix E: Calibration Rationale and Sensitivity

The 30/70 portfolio split and the 0.35/0.15 unsafe probability parameters are plausible starting values grounded in published evidence, but their precise calibration has not been validated against observed institutional data. To address this, the threshold sensitivity analysis (below) tests the robustness of findings across a wide parameter range. Additionally, the simulation specification document (Multimedia Appendix 2) recommends supplementary calibration sensitivity runs varying the portfolio composition (20/80 through 50/50) and unsafe probabilities ($\pm 30\%$ of default values) to establish whether the primary findings are robust to alternative calibration assumptions. These supplementary analyses are designed to be run without prejudging results.

Supplementary Appendix F: Parameter and Evidence Sensitivity as Formal Experimental Axes

Two formal research axes emerge from the gate architecture and merit explicit identification as the empirical programme that must follow this simulation contribution. Crucially, these axes constitute a two-dimensional empirical problem rather than separate lines of inquiry. Interaction is defined operationally as non-additivity of outcome sensitivity under joint perturbation of threshold parameters and evidence confidence distributions. In this context, an interaction effect is present where the combined influence of parameter variation and evidence variation on deployment decisions differs from the sum of their independent effects. Formal characterisation of this interaction is a prespecified objective for future empirical work using factorial simulation designs and variance-based sensitivity analysis (e.g. Sobol interaction indices).

Governance behaviour is best understood as a decision surface defined jointly over parameter space and evidence space, where outcomes depend on the interaction between threshold calibration and evidence quality rather than on either dimension independently. The first is parameter sensitivity: the gate thresholds, override validity periods, concurrent override limits, and escalation triggers are provisional design defaults whose optimal values are an empirical question. The companion simulation study indicates that the safety gate threshold is the dominant governance lever; small changes in this parameter produce non-linear effects on deployment outcomes, including cliff-edge behaviour where a 0.05-unit shift reverses deployment decisions. The second is evidence sensitivity: the quality, provenance, and completeness of governance evidence determine whether gates are satisfied, and these properties vary substantially across cases and institutions. A governance architecture that is insensitive to evidence quality would be undesirable—it would approve poorly documented systems as readily as well-documented ones. The sensitivity of the gate architecture to both parameter calibration and evidence quality is therefore a designed feature, not a limitation, provided that both dimensions are governed transparently. A governance architecture that is insensitive to evidence quality or threshold calibration would be normatively undesirable, as it would produce invariant decisions under epistemically distinct conditions. Sensitivity to both

parameterisation and evidence quality is therefore a necessary property of a governance architecture intended to reflect underlying risk and uncertainty. The pilot design proposed above is structured to characterise both axes under institutional conditions. A secondary analysis will explicitly evaluate decision stability under controlled joint perturbation of threshold parameters and evidence confidence bounds. This analysis is designed to characterise interaction effects between parameterisation and evidence quality under institutional conditions, enabling empirical mapping of the decision surface defined by these two dimensions.

Decision Analysis Context

In safety-critical systems engineering, Leveson's systems-theoretic accident model [9] and barrier-based safety analysis both formalise the principle that safety constraints must not be compensated against each other. The present study extends this logic to clinical AI governance by simulating the comparative safety properties of conjunctive versus compensatory decision rules under controlled evidence distributions. Its contribution is not the conjunctive principle itself—which is well established in both traditions—but its application to and empirical characterisation within the specific institutional context of clinical AI deployment decisions.

Relationship to FUTURE-AI and Existing Governance Frameworks

The FUTURE-AI consensus guideline [7] defines six guiding principles with 30 best practices spanning the AI lifecycle. These principles address what must be governed but do not specify the institutional decision rule for converting governance evidence into a deployment decision.

Supplementary Appendix K: Epic Sepsis Model Case Scenario

Case Scenario: Epic Sepsis Model Applied to the sepsis model scenario, the gate architecture produced a NO-GO decision. Under the high-risk tier thresholds (G1: 0.65, G2–G5: 0.60), the case failed all five gates: G1 ($0.58 < 0.65$), G2 (missing, treated as fail), G3 ($0.38 < 0.60$), G4 ($0.52 < 0.60$), and G5 ($0.30 < 0.60$). The gate architecture provides immediate transparency about which specific domains drove the refusal and therefore which remediation steps are required. Under composite scoring, the deployment outcome depends on the threshold applied and the imputation method for missing data. With mean imputation for the missing equity score (G2), the weighted composite score is 0.52—below both the moderate and matched thresholds, producing a NO-GO. However, the composite architecture does not identify which domains failed or which specific evidence gaps produced the aggregate shortfall. Under gate logic, the monitoring failure (G5 = 0.30) is immediately visible as the widest gap, directing remediation effort toward alert-fatigue threshold specification and monitoring infrastructure.

Gate Failure Patterns

Under the primary model, failures were most frequent in monitoring (Gate 5) and equity (Gate 2) domains, confirming that heterogeneous evidence profiles created latent-domain vulnerabilities that conjunctive logic captures. This pattern is consistent with the design of the heterogeneous model, in which Gates 2 and 5 receive the largest adverse shifts in latent evidence quality for unsafe tools.

Supplementary Appendix O: Empirical Heterogeneity — The Key Testable Prediction

The simulation's central testable prediction is that governance evidence heterogeneity — where unsafe AI tools exhibit systematically deficient evidence in less-visible domains (equity, monitoring) than in audit-visible domains (safety, documentation) — is common in practice. Preliminary evidence from the historical replay study supports this prediction: among 12 documented failures, gate failure patterns were heterogeneous, with 83% failing the safety gate but only 25% failing the evidence gate, consistent with the model's assumption that evidence visibility varies across governance domains. Formal empirical testing requires collecting governance evidence profiles from institutional AI portfolios and assessing whether the heterogeneous pattern predominates.

Supplementary Appendix Q: Counterfactual Analysis of Composite Threshold Calibration

If composite thresholds were fixed ex ante by independent safety committees at levels calibrated to prevent unsafe deployment—rather than adjusted under institutional pressure to maximise deployment volume—the composite method could achieve equivalent safety to gates, as the matched-threshold analysis indicates. The gate advantage documented in this simulation is therefore not purely mathematical; it is a governance result. Conjunctive logic structurally prevents the threshold erosion that institutional deployment pressure produces in composite systems, because relaxing one gate threshold does not permit compensation from other domains.

Framing as Theory-Testing and the Research Agenda for Pilots

This study is a theory-testing simulation, not an empirical study of AI governance practice. Its value lies in three contributions: (a) demonstrating that the theoretical mechanism—compensation prevention by conjunctive logic—operates as predicted under the specified model; (b) identifying the scope condition under which it does not operate; and (c) generating testable predictions for real-world pilot studies.

Supplementary Appendix T: Epic Sepsis Case Scenario Evidence Score Assignments

Domain evidence scores were assigned on the basis of documented characteristics: strong vendor-reported performance metrics (G1 partial), documented equity evaluation gaps (G2 fail), proprietary documentation constraints (G3 fail), variable withdrawal authority documentation (G4 partial), and absent alert-fatigue thresholds (G5 fail). This scenario illustrates how the governance architecture would process a set of evidence scores in practice; it is not a formal empirical audit of the named system, and the evidence scores are the author's interpretive assignments based on published literature rather than verified institutional records.

Results

Two safety metrics are reported throughout and should be distinguished. The unsafe deployment rate is the proportion of all tools in the portfolio that are both deployed and genuinely unsafe—an absolute population-level risk measure. The contamination rate (unsafe proportion among deployed tools) measures the probability that any individual deployed tool is unsafe—a per-deployment risk visible to clinicians and patients.

This work forms part of a coordinated multi-paper research programme examining non-compensatory governance architectures across formal, empirical, and operational domains.

Multimedia Appendix 3. Generative AI Use Log

No AI tool was used to design the simulation architecture, select parameters, interpret results, specify scope conditions, or draft the analytical arguments and conclusions. All simulation design decisions, parameter choices, interpretation, and textual content were produced by the author. A complete prompt/response log is retained and can be supplied to editors and reviewers on request (Multimedia Appendix 3), consistent with JMIR's editorial policy on generative AI use. No patient-identifiable information or confidential organisational information was entered into any external AI tool. The author reviewed and verified all content and references and takes full responsibility for the manuscript. Conflicts of Interest: WB is the founder of Ethical Alpha Audit, which develops governance methodologies based on this gate architecture. WB is also an analyst at NHS England; these views are the author's own and do not represent NHS England. Given this commercial interest, independent replication of the simulation findings is explicitly encouraged and the repository's Quick Start guide is provided specifically to facilitate this.

Generative AI use log: prompts, responses, and change-tracking summary. Abbreviations AI: artificial intelligence; AIDRS: AI and Digital Regulations Service; CAIOC: Clinical AI Oversight Committee; CI: confidence interval; DTAC: Digital Technology Assessment Criteria; FDA: Food and Drug Administration; MHRA: Medicines and Healthcare products Regulatory Agency; NHS: National Health Service; NICE: National Institute for Health and Care Excellence; SaMD: software as a medical device; SD: standard deviation; SIRO: Senior Information Risk Owner